

Python Built-in Modules Handbook

This handbook covers essential built-in modules for interviews and projects.

math

Purpose: Mathematical functions

Common functions: `sqrt()`, `pow()`, `factorial()`, `ceil()`, `floor()`, `pi`

Example:

```
import math
```

random

Purpose: Random values

Common functions: `randint()`, `choice()`, `shuffle()`, `random()`

Example:

```
import random
```

datetime

Purpose: Dates and time

Common functions: `datetime.now()`, `date.today()`, `timedelta`

Example:

```
import datetime
```

os

Purpose: Operating system

Common functions: `getcwd()`, `listdir()`, `makedirs()`, `remove()`, `rename()`

Example:

```
import os
```

sys

Purpose: Python runtime

Common functions: `argv`, `path`, `version`, `exit()`

Example:

```
import sys
```

re

Purpose: Regular expressions

Common functions: `search()`, `match()`, `findall()`, `sub()`

Example:

```
import re
```

json

Purpose: JSON

Common functions: `load()`, `loads()`, `dump()`, `dumps()`

Example:
`import json`

CSV

Purpose: CSV

Common functions: `reader()`, `writer()`

Example:
`import csv`

collections

Purpose: Advanced containers

Common functions: `Counter`, `defaultdict`, `deque`

Example:
`import collections`

itertools

Purpose: Iterators

Common functions: `count()`, `cycle()`, `combinations()`, `permutations()`

Example:
`import itertools`

functools

Purpose: Functional

Common functions: `reduce()`, `lru_cache()`

Example:
`import functools`

statistics

Purpose: Statistics

Common functions: `mean()`, `median()`, `mode()`

Example:
`import statistics`

pathlib

Purpose: Paths

Common functions: `Path`

Example:
`import pathlib`

shutil

Purpose: Files

Common functions: `copy()`, `move()`, `rmtree()`

Example:
`import shutil`

subprocess

Purpose: External commands

Common functions: `run()`, `Popen()`

Example:
`import subprocess`

threading

Purpose: Threads

Common functions: `Thread()`, `Lock()`

Example:
`import threading`

logging

Purpose: Logs

Common functions: `basicConfig()`, `info()`, `error()`

Example:
`import logging`

sqlite3

Purpose: Database

Common functions: `connect()`, `cursor()`

Example:
`import sqlite3`